Problem Set 2.1 (Bonus) – Climatic Risk Intelligence Module

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Source Code

```python
"""
Program: Climatic Risk Intelligence Module
Purpose: Classifies a work-site into a safety tier (Freeze Alert, Critical,
         Cautionary, Operational) from a derived Heat Stress Index, using
         the walrus operator for input capture and short-circuit logic.
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Heuristic Tiering Model: HSI = Temperature + (0.5 * Humidity). Freeze
Alert is checked first regardless of HSI, since a high-humidity freezing
site can still produce a numerically large HSI. Critical and Cautionary
are checked next in that order; Operational is the catch-all else, which
avoids an unnecessary explicit lower-bound condition.
"""

if (temp_str := input("Temperature (C): ").strip()):
    try:
        temperature = float(temp_str)
    except ValueError:
        temperature = None
else:
    temperature = None

if temperature is None:
    print("Safety Level: Unknown")
else:
    humidity_str = input("Humidity (%): ").strip()
    try:
        humidity = float(humidity_str)
        assert humidity >= 0, "Telemetry Error: Negative Humidity"
    except ValueError:
        humidity = None

    if humidity is None:
        print("Safety Level: Unknown")
    else:
        hsi = temperature + (0.5 * humidity)

        if temperature <= 0:
            tier = "FREEZE ALERT"
        elif hsi > 45 or (temperature > 38 and humidity > 70):
            tier = "CRITICAL"
        elif 30 <= hsi <= 45:
            wind_str = input("Wind speed (km/h): ").strip()
            wind_speed = float(wind_str) if wind_str else 0.0
            tier = "CAUTIONARY" if wind_speed < 5 else "OPERATIONAL"
        else:
            tier = "OPERATIONAL"

        # Challenge Task: Nested Intelligence Suite (battery adjustment)
        if tier == "CAUTIONARY":
            battery_str = input("Battery level (%): ").strip()
```

```
            battery = float(battery_str) if battery_str else 100.0
            if battery < 20:
                tier = "CRITICAL"
            elif battery > 80:
                tier = "OPERATIONAL"

        risk_label = "Safe" if hsi < 30 else "Unsafe"
        print(f"HSI: {hsi:.1f}")
        print(f"Safety Level: {tier}")
        print(f"Risk Label: {risk_label}")
```

Execution Trace

```
--- Scenario (a): Empty input handled by walrus check ---
Temperature (C):
Safety Level: Unknown

--- Scenario (b): Critical heatwave reading ---
Temperature (C): 32
Humidity (%): 80
HSI: 72.0
Safety Level: CRITICAL
Risk Label: Unsafe

--- Scenario (c): Cautionary reading adjusted by Battery-Level bonus ---
Temperature (C): 25
Humidity (%): 20
Wind speed (km/h): 3
Battery level (%): 10
HSI: 35.0
Safety Level: CRITICAL
Risk Label: Unsafe

Note: primary tier for (25, 20, wind 3) is CAUTIONARY (30 <= HSI <= 45,
wind < 5). Battery level 10 is below the 20% threshold, so the Nested
Intelligence Suite elevates it to CRITICAL.
```

Submission Notes

Freeze Alert is checked before anything else because temperature alone decides it – a freezing site with high humidity can still produce a large HSI, and checking HSI-based tiers first would misclassify it as a heat emergency. Critical and Cautionary come next in that order, so by the time a Cautionary branch runs it already knows HSI is not above 45.

The empty-input check runs on the raw string from `input()`, guarded by the walrus operator, before any conversion happens. This matters because a temperature of `0` is falsy as a number but not falsy as the string `"0"`, so checking the raw string avoids treating a genuine 0°C reading as missing input.